

# The rank-one map $\ell^1 \rightarrow \ell^\infty$ in arXiv:2603.01336 is disjoint weak-star Dunford-Pettis

## Source passage

After Definition 3.2, Boumnidel and Hafidi consider the operator

$$T : \ell^1 \longrightarrow \ell^\infty, \quad T((\lambda_k)_{k=1}^\infty) = \left( \sum_{k=1}^\infty \lambda_k, \sum_{k=1}^\infty \lambda_k, \dots \right),$$

and state that this  $T$  is not disjoint weak-star Dunford-Pettis. The relevant source pages are reproduced in Figures 1 and 2.

## Claim

**Theorem 1.** *Let  $T : \ell^1 \rightarrow \ell^\infty$  be defined by*

$$Tx = \left( \sum_{k=1}^\infty x_k \right) \mathbf{1}, \quad \mathbf{1} = (1, 1, \dots).$$

*Then  $T$  is disjoint weak-star Dunford-Pettis.*

*Proof.* Let  $(x_n)$  be a norm-bounded disjoint sequence in  $\ell^1$ , and let  $(f_n) \subset (\ell^\infty)'$  satisfy  $f_n \rightarrow 0$  in the weak-star topology  $\sigma((\ell^\infty)', \ell^\infty)$ . Put

$$\sigma(x) = \sum_{k=1}^\infty x_k, \quad x \in \ell^1.$$

Then  $\sigma \in (\ell^1)'$ , so the scalar sequence  $(\sigma(x_n))$  is bounded. Also, since  $\mathbf{1} \in \ell^\infty$  and  $f_n \rightarrow 0$  weak-star,

$$f_n(\mathbf{1}) \longrightarrow 0.$$

Therefore

$$f_n(Tx_n) = f_n(\sigma(x_n)\mathbf{1}) = \sigma(x_n)f_n(\mathbf{1}) \longrightarrow 0.$$

This is precisely the definition of a disjoint weak-star Dunford-Pettis operator. □

**Remark 1.** *The displayed calculation in the source cannot produce a counterexample: any weak-star null sequence in  $(\ell^\infty)'$  must vanish on the constant vector  $\mathbf{1}$ . In particular, coordinate functionals  $e_n$  are not weak-star null in  $(\ell^\infty)'$ , since  $e_n(\mathbf{1}) = 1$  for every  $n$ . The argument above only refutes the printed assertion about this particular rank-one operator; it does not address whether some different example can serve the intended purpose.*

Since  $(\pi_{k_n} \circ T)$  is  $w^*$ -null sequence, by [[8], Lemma 2.2] we have  $(f_n)$  is  $w^*$ -null sequence in  $E'$ . By hypothesis, it follows from Definition 3.1 that  $f_n(x_n) \rightarrow 0$  as  $n \rightarrow \infty$ . On the other hand,

$$|f_n(x_n)| = |(\pi_{k_n} \circ T)(x_n)| = |\pi_{k_n}(T(x_n))| = T(x_n) > \epsilon > 0.$$

This is a contradiction. Hence,  $\|T(x_n)\| \rightarrow 0$  for every disjoint sequence  $(x_n)$  of  $E$ , that is,  $T$  is an M-weakly compact operator.

(2)  $\Rightarrow$  (1) By Definition 3.1, to prove that  $E$  has the disjoint weak\* Dunford-Pettis property, we need to prove that for each norm bounded disjoint sequence  $(x_n)$  of  $E$  and each disjoint  $w^*$ -null sequence  $(f_n) \subset E'$ , we have  $f_n(x_n) \rightarrow 0$ . Let consider the operator  $T : E \rightarrow c_0$  defined by  $T(x) = (f_n(x))_n$  for any  $x \in E$ . By hypothesis,  $T$  is M-weakly compact. Therefore,  $\|T(x_n)\| \rightarrow 0$  as  $n \rightarrow \infty$ , hence  $f_n(x_n) \rightarrow 0$ , as desired.

(2)  $\Leftrightarrow$  (3) By virtue of [[17], Theorem 18]. □

**Proposition 3.3.** *Let  $E$  be a Banach lattice with the disjoint weak\* Dunford-Pettis\* property, If  $E'$  and  $E$  are order continuous, then  $E$  is finite dimensional.*

*Proof.* Let  $(x_n)$  be a disjoint norm bounded sequence of  $E$ , by Lemma 2 and Lemma 1 of [16],  $|x_n| \xrightarrow{uaw} 0$ , then by [[16], Theorem 7] we have  $|x_n| \xrightarrow{w} 0$ . (\*)

Let  $(f_n)$  be a norm bounded disjoint sequence in  $E'$ , by [10, Corollary 2.4.3],  $f_n$  is  $w^*$ -null sequence, by assumption,  $f_n(x_n) \rightarrow 0$ . (\*\*)

From (\*) and (\*\*) and [[4], Corollary 2.6], we conclude that  $x_n \xrightarrow{\|\cdot\|} 0$ .

Now, by [[17], Theorem 18] and [[11], Corollary 5] we claim that  $E$  is finite dimensional. □

We introduce the weak\* version of uaw-Dunford-Pettis operators.

**Definition 3.2.** *An operator  $T : E \rightarrow Y$  from a Banach lattice  $E$  into a Banach space  $Y$  is said to be disjoint weak\* Dunford-Pettis, if for every norm bounded disjoint sequence  $(x_n)$  in  $E$ , and every sequence  $f_n \xrightarrow{w^*} 0$  in  $Y'$  imply,  $\lim_{n \rightarrow \infty} f_n(T(x_n)) = 0$ .*

Now, we consider some ideal properties for the class of disjoint weak\* Dunford-Pettis operator. But before investigating this, we give the following Example.

$$\begin{aligned} T : \ell^1 &\longrightarrow \ell^\infty \\ \lambda_n &\longmapsto \left( \sum_{n=1}^{\infty} \lambda_n, \sum_{n=1}^{\infty} \lambda_n, \dots \right) \end{aligned}$$

Figure 1: Definition 3.2 and the displayed rank-one operator.

By Remark 3.3 and Proposition 3.7  $Id_{\ell^\infty}$  is a disjoint weak\* Dunford-Pettis operator, However  $T = Id_{\ell^\infty} \circ T$  is not disjoint weak\* Dunford-Pettis. Indeed, let  $f_n \xrightarrow{w^*} 0$  in  $(\ell^\infty)'$ , then there exists a unique sequence of scalars  $(\alpha_n)$  such that  $f_n = \sum_{n=1}^{\infty} \alpha_n e_n$ . Moreover, we have that

$$\begin{aligned} \langle T(e_n), f_n \rangle &= \langle (\sum_{n=1}^{\infty} e_n, \sum_{n=1}^{\infty} e_n, \dots), \sum_{n=1}^{\infty} \alpha_n e_n \rangle \\ &= \sum_{n=1}^{\infty} \alpha_n \langle (\sum_{n=1}^{\infty} e_n, \sum_{n=1}^{\infty} e_n, \dots), e_n \rangle \\ &= \sum_{n=1}^{\infty} \alpha_n \cdot \sum_{n=1}^{\infty} e_n \langle (1, 1, \dots), e_n \rangle \\ &= \sum_{n=1}^{\infty} \alpha_n \cdot \sum_{n=1}^{\infty} e_n \rightarrow 0 \end{aligned}$$

We clearly see that  $T$  is not a disjoint weak\* Dunford-Pettis operator. We also claim that  $T$  is not disjointness preserving operator. Indeed, since  $(e_n)$  is disjoint then  $T(e_n \wedge e_m)$  is null in  $\ell^\infty$ , but

$$T(e_n) \wedge T(e_m) = (1, 1, 1, \dots) \neq (0, 0, 0, \dots)$$

**Proposition 3.4.** *Let  $S : E \rightarrow F$  and  $T : F \rightarrow G$  be two operators between Banach lattices  $E, F$  and  $G$ .*

- (1) *If  $T$  is bounded and  $S$  is a disjoint weak\* Dunford-Pettis operator, then  $T \circ S$  is also disjoint weak\* Dunford-Pettis.*
- (2) *If  $S$  is disjointness preserving and  $T$  is a disjoint weak\* Dunford-Pettis operator, then  $T \circ S$  is disjoint weak\* Dunford-Pettis.*

*Proof.* (1) Let  $(x_n)$  be a norm bounded disjoint sequence of  $E$  and  $(f_n) \subset G'$  such that  $f_n \xrightarrow{w^*} 0$ . Then  $f_n \circ T \xrightarrow{w^*} 0$   $(\star)$ .

By assumption  $S$  is disjoint weak\* Dunford-Pettis, it follows from  $(\star)$  that  $f_n \circ T(S(x_n)) = f_n(T \circ S(x_n)) \rightarrow 0$ , for each null weak\* convergent sequence  $(f_n)$  in  $G'$ . Hence  $T \circ S$  is disjoint weak\* Dunford-Pettis.

(2) Let  $(x_n)$  be a disjoint sequence of  $E$  and  $(f_n) \subset G'$  such that  $f_n \xrightarrow{w^*} 0$ . As,  $S$  is disjointness preserving operator, it follows that  $(S(x_n))$  is a disjoint sequence. In this step, assumption asserts that  $f_n(T(S(x_n))) \rightarrow 0$ . Therefore,  $T \circ S$  is disjoint weak\* Dunford-Pettis.  $\square$

Figure 2: The source passage claiming that  $T$  is not disjoint weak-star Dunford-Pettis.

## Human Review: rank-one map $\ell^1 \rightarrow \ell^\infty$

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Not independently checked in detail by the human reviewer; excluded from the survey.

Verdict: Excluded from the main survey.

## Human Comments

The packet claims to identify a local error in the source paper. The paper states that the rank-one operator

$$T : \ell^1 \rightarrow \ell^\infty, \quad T(x) = \left( \sum_{k=1}^{\infty} x_k \right) \mathbf{1}, \quad \mathbf{1} = (1, 1, \dots),$$

is not disjoint weak-star Dunford-Pettis. The automatic packet argues that this assertion is false.

Indeed, write

$$\sigma(x) = \sum_{k=1}^{\infty} x_k.$$

Then

$$T(x) = \sigma(x) \mathbf{1}.$$

If  $(x_n)$  is a norm-bounded disjoint sequence in  $\ell^1$ , then  $(\sigma(x_n))$  is bounded. If

$$f_n \xrightarrow{w^*} 0 \quad \text{in } (\ell^\infty)',$$

then, since  $\mathbf{1} \in \ell^\infty$ ,

$$f_n(\mathbf{1}) \longrightarrow 0.$$

Therefore

$$f_n(Tx_n) = f_n(\sigma(x_n) \mathbf{1}) = \sigma(x_n) f_n(\mathbf{1}) \longrightarrow 0.$$

This would be exactly the definition of a disjoint weak-star Dunford-Pettis operator, and hence would refute the displayed assertion in the source paper.

The alleged mistake in the source paper appears to come from an incorrect handling of weak-star null sequences in  $(\ell^\infty)'$ . In particular, one cannot generally represent arbitrary elements of  $(\ell^\infty)'$  by countable coordinate sums, and any weak-star null sequence in  $(\ell^\infty)'$  must vanish on the constant vector  $\mathbf{1}$ .

## Short Version

The automatic packet gives a short argument that the rank-one map factors through the constant vector  $\mathbf{1}$ , and that weak-star null functionals on  $(\ell^\infty)'$  vanish on  $\mathbf{1}$ . If correct, this would show that the operator is disjoint weak-star Dunford-Pettis, contrary to the assertion in the source paper.

## Literature and Follow-Up

This should be treated, at most, as a possible local correction to a printed example, not as a substantive new mathematical discovery. The surrounding paper also shows some signs of technical unreliability, so this packet is best understood as an example-checking case rather than as progress on an open problem.

## Review Outcome

Exclude this packet from the main survey of successful mathematical discoveries. It may be retained as a cautionary example of a possible local paper-error detection, but the argument has not been independently checked in detail by the human reviewer, and in any case the mathematical contribution would be too small and too local to count as a main survey example.